Draft ER390 Project Proposal

Visualizing HAT's GOE Monitoring Data

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The Idea behind submitting this project proposal

After hearing the suggestion to create a data visualization tool for HAT GOE Monitoring as a suitable ER390 Final Project, I began thinking about several creative solutions. I gathered data from the project publication that influenced HAT's project design, began to develop coding ideas, modifiable to visualize HAT data when available in October, which could also prove useful to develop a framework to compare results between multiple Garry Oak Ecosystem Monitoring Projects if similar assessment measurements and indices have been used (Malloff & Shackelford, 2024). See *Proposed Gantt Timeline Chart*

Background

Comparing monitoring results from multiple restoration projects requires consistent measurements of vegetation composition (e.g. native and exotic species cover), which may help identify responses to stressors affecting already compromised and rare Garry Oak Ecosystems. Representing data in a visual manner can help make complex data more understandable than a tables of numbers, and may reveal patterns or trends resulting from different restoration management actions (Malloff & Shackelford, 2024). Data visualization can represent abstract information visually, by allowing for data exploration and analysis, modelling of restoration management alternatives, data-based story-telling, and communicating potential insights, to help understand and make restoration management decisions. Using RMarkdown to generate reports helps create reproducible research methods (McKay, 2020).

° Malloff, J., & Shackelford, N. (2024). Feeling the Pulse: Monitoring methods and initial outcomes in oak meadow ecosystems. Restoration Futures Lab at the University of Victoria.

° McKay, S. K. (2020). Data visualization for ecological analysis and restoration. US Army Corp of Engineers:Ecosystem Management and Restoration Research Program (EMRRP) Webinar Series. https://cw-environment.erdc.dren.mil/webinars/20Feb5-EcoMod-DataViz.pdf

Project Partner

I spoke to Vanessa Brownlee, Habitat Restoration Coordinator at Habitat Acquisition Trust (HAT) Social on 2025-08-27, who expressed interest in my proposed project idea, and seemed willing to consider letting me start in Fall 2025, with data available later in October. I also later spoke briefly to Board Chair Andy MacKinnon, who seemed keen to know of my interest in doing my RNS Final Project with HAT and looks forward to learning more. I'm prepared to be guided by HAT's preferences, timing, and data protocols.

Project Goals and Overall Purpose (problem/opportunity)

I propose to create a customized, coding and reporting tool, including tutorial instructions, for use by non-coders, using open-source R package libraries, for data formatting and interactively visualizing data collected at multiple sites for HAT Garry Oak Ecosystem (GOE) Monitoring Program, to provide an interactive way to communication information about the data, and help determine any patterns through analysis. This project proposal was created using R code in RMarkdown, to showcase some visualizations tools and reports that could be created for HAT, after adapting code to reflect available HAT data. An overarching goal would be to create code to compare GOE Monitoring Projects which use the same measurement parameters.

Project Objectives

- See *WorkFlowchart, Figure 1*, and Gantt Chart *Proposed Timeline, Figure 2*
- **Analyze and transform collected data** to understand underlying structure, ensure use of consistent variable naming protocols, format data to use with code, calculate metrics.
- **Write data visualization code** for HAT data using RStudio, RMarkdown, and RShiny, using reproducible code to create appropriate visualizations to enable a better understanding of the collected data,

for use by non-coders, to automatically download and format data, and produce customized reports. Create custom R functions to read, analyze, and plot data, as basis for HAT R package.

- **Create interactive data visualization tools** Scaleable, allow for more sites added, multi-year data, different measuring and sensitivity criteria (e.g. First Nations GOE sites); customize for consistent styles using HAT colours, logos, and plot themes.
- **Create interactive, browser-based Shiny App** User-chosen, custom-make data visualization plots, maps, and statistical analysis; select location(s) and variables; download results.
- **Develop code to compare GOE Monitoring Projects Data** HAT data with ECCC data using same measurement parameters (Malloff & Shackelford, 2024),
- **Analyze results for patterns** Track individual site differences using variables and other metrics/indices that HAT measures.

Project Methods for HAT Data visualization

- See *WorkFlowchart, Figure 1*, and Gantt Chart *Proposed Timeline, Figure 2*
- Continue research for Shiny reactive code to make interactive data choices (e.g. drop-down lists), respond to choices, filter results. Interactive choices will eventually react to other variable selections to make custom-made visualizations and analysis with only a few clicks
- Sign any HAT data confidentiality agreements as may be required
- Consult with HAT restoration monitor team to explore and understand the nature of the collected data, data format (forms, spreadsheets), the variables that are being measured, indices to be calculated, questions HAT wants to explore with the data to analyze resuts
- Create a portable Project folder, with subfolders for R scripts, raw data, processed data, images, reports, meta data (units, data descriptions, etc); create file and variable naming protocols
- Analyze and clean data, check for errors, inconsistencies; calculate indices values from raw data
- Create data visualization plot charts that appropriately reflect nature of variables being measured
- Document methods, successes, and failures in the process of the project for final paper
- Continued research for using R to create reproducible code, and effective use of data visualization in ecological restoration monitoring, for inclusion in final report
- Develop functional parts one-at-a-time to ensure correct operation before starting next part as coding is not always a linear process, with some functions taking longer to develop; troubleshoot small sections to get a fully working product before moving further adding more functionality

Proposed Deliverables

See *WorkFlowchart, Figure 1*, and *Proposed Timeline, Figure 2*; All deliverables created using RStudio
Data Visualizations: Including data tables, plots, interactive maps; for subgroups, individual sites, site comparison, or for confidential use by First Nations; See *Example Visualization*
RMarkdown Data Processing Document: Data downloading, data cleaning and preparation for use in visualizations, create data visualizations for use in Shiny App; Create reports in HTML, pdf, or Word with plot output images, or code snippets
Customizations: Create customized colour palette for Plot themes, App, Reports; add HAT logo
Shiny App Interactive Data Viz Tool: Data processed in RMarkdown document; choose data variables to visualize tables, plot, maps, stats; run local for confidentiality *https://goe-interact.shinyapps.io/Test-Shiny-GOE-Monitor/*
Tutorials for using Data visualization Tools: customized, easy to understand
Create Report Templates: Customized fpr individual site, total project reports, or as requested by HAT
Best practises for data files: To ensure naming consistency in files, variables, folder structure, etc.
Create Customized Functions for HAT to form core of a HAT-specific R Package library
Presentation: Completed project presented to community partner HAT; including slide show, live demonstrations of Data Visualization Tools, and final copy of my written report when approved to be published.
Written Report: Due 2026-08-15, document project, using R code, and visualizing ecological restoration monitoring data; publish in Ecorestoration: RNS Technical Series

Workflow Graph

- See *Objectives, Methods, and Deliverables*
- The workflow shows how the raw data, from HAT monitoring measurements, would be transformed and included in code to produce a variety of deliverable products.
- This graph is made with code from DiagrammeR, using a colour-blind friendly colour palette.

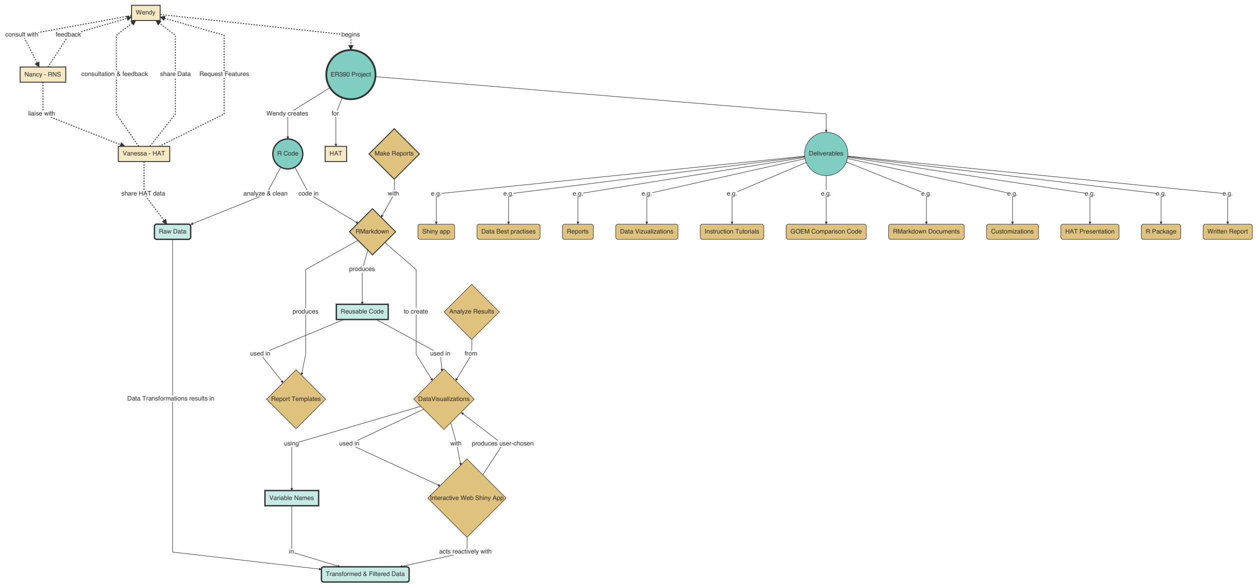

Figure 1: Project method workflow diagram for ER390 data visualization for HAT

Proposed Timeline

- red coloured items are considered critical
- grey items are completed
- Note: This chart was coded with DiagrammeR using an imported csv file of data, however, there is no colour control
- See *Objectives, Methods, and Deliverables*

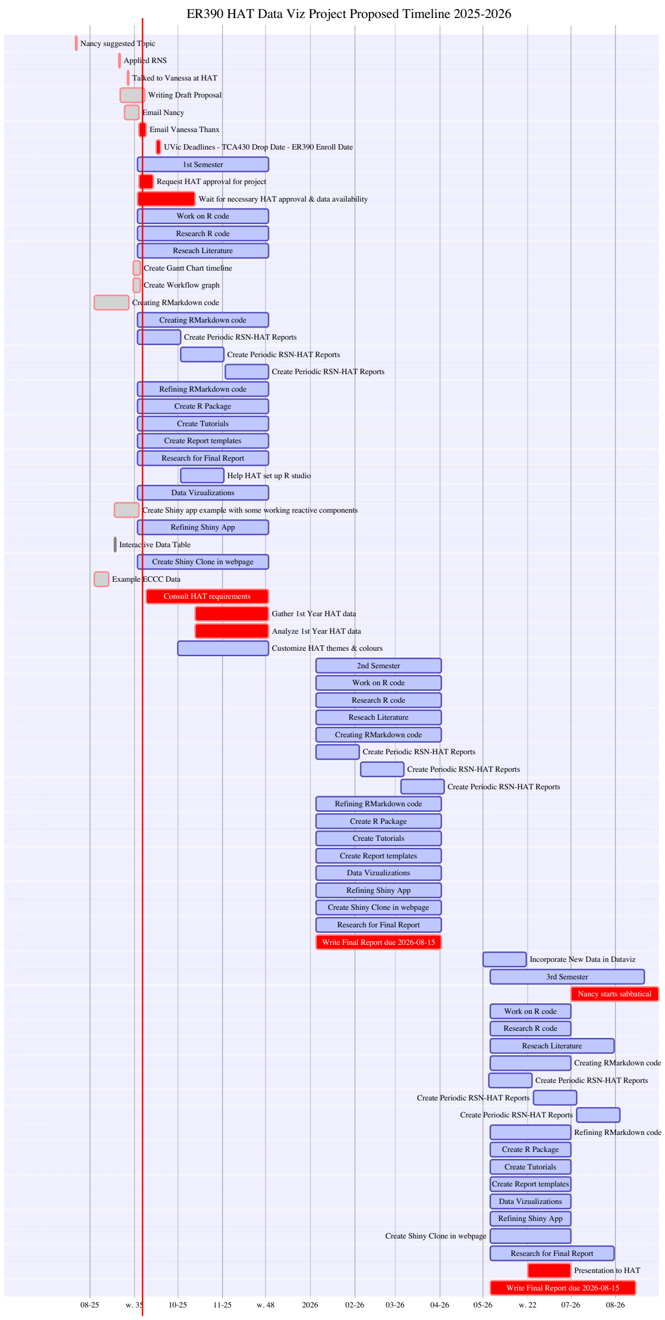

Figure 2: Gantt Proposed ER390 Timeline

Example Data Visualization Results

Data download and formatting

- These example visualizations use GOE monitoring data from report by Malloff & Shackelford (2024)
- Data (in excel or csv format) is uploaded
- Data is formatted, saved, and managed to be useful for creating visualizations
- The code has been hidden for the HTML report, highlighting only the resulting plots, though is visible in PDF document. The R code has been documented using hidden comments.

Load Site Data

Click open to see R code for `Loading Data`

```
# Load data and create 2 data frames
# Load Site Data
geo_eccc <- read.csv("data/SiteData-TEST-GOE-Monitor.csv", header = TRUE, sep = ",")
# Load Site Details
Sites_eccc <- read.csv("data/SiteDetails-TEST-GOE-Monitor.csv", header = TRUE, sep = ",")
```

Data Cleaning and Transformation

Click open to see R code for `Data Cleaning and Transformation`

```
# create new column with proportions changed to percentage as a whole number
Percentage_ns <- geo_eccc$Proportion_of_native_species * 100

# add-new column for Percentages
geo_eccc <- cbind(geo_eccc, Percentage_ns)

# arrange order of site names
Sites_eccc_1 <- Sites_eccc %>% arrange(Sites_eccc$Site)
geo_eccc_1 <- geo_eccc %>% arrange(geo_eccc$Site)

# bind 2 dataframes together
# this requires Sites rows from both data frames to be in alphabetical order
## to assign the data rows correctly
geo_eccc_sites <- cbind(Sites_eccc_1, geo_eccc_1)

# Remove duplicate columns resulting from binding 2 data frames
geo_eccc_sites_1 <- geo_eccc_sites[c(-9,-10)]

# Rename columns to remove dot-format
names(geo_eccc_sites_1)[names(geo_eccc_sites_1) == "Land.Manager"] <- "Land_Manager"
names(geo_eccc_sites_1)[names(geo_eccc_sites_1) == "Main.Restoration.Type"] <- "Main_Restoration_Type"
names(geo_eccc_sites_1)[names(geo_eccc_sites_1) == "Restoration.Intensity"] <- "Restoration_Intensity"
names(geo_eccc_sites_1)[names(geo_eccc_sites_1) == "Area.ha"] <- "Area_ha"

# write csv file with combined data
## to create new data file which is used for creating the visualizations
write.csv(geo_eccc_sites_1, "data/geo_eccc_all_site_data.csv", row.names = FALSE)
```

Data Visualization Example Results

Data Table

- using data from Malloff & Shackelford (2024)

Full Data Table

- The full table is too wide for use in pdf because current variable names are too long.

Subregion Site Land_Manager Main_Restoration_Type Restoration_Intensity Area_ha Lat Lng Proportion_of_native_species Cultural_species_richness Exotic_species Trampling Herbivory Composite_Index Year Percentage_ns

Table Head of Site Information

Subregion	Site	Land.Manager	Main.Restoration.Type	Restoration.Intensity	Area.ha	Lat	Lng
Gulf Islands	Anniversary Island	GINPR	herbivore reduction	high	4.39	48.82292	-123.1823
Gulf Islands	AVNR	Saltspring conservancy	herbivore reduction	minimal	20.54	48.80351	-123.4425
Saanich Peninsula	Bear Hill Park	CRD	invasive removal	low	3.80	48.54639	-123.4078
Gulf Islands	Brackman Island	GINPR	invasive removal	high	4.41	48.71897	-123.3864
Saanich Peninsula	Camas Hill	HAT	invasive removal	high	10.10	48.40173	-123.5975
Gulf Islands	Crows Nest	Trinity Western University	herbivore reduction	high	15.34	48.78237	-123.4612

Table Head of Site Data

Subregion	Site	Proportion_of_native_species	Cultural_species_richness	Exotic_species	Trampling	Herbivory	Composite_
Gulf Islands	Anniversary Island	0.94	2.71	0.17	5.00	1.14	
Gulf Islands	AVNR	0.55	1.30	57.19	4.70	36.50	
Saanich Peninsula	Bear Hill Park	0.62	1.86	24.46	7.57	9.50	
Gulf Islands	Brackman Island	0.77	2.80	11.17	5.40	0.00	
Saanich Peninsula	Camas Hill	0.61	2.43	27.19	7.71	4.71	
Gulf Islands	Crows Nest	0.66	1.07	31.51	1.64	26.71	

Figures 3,4,5: Example Data Tables

Map of Site Locations

Leaflet Map with Circle Markers

- The Leaflet Map is created using R code in RMarkdown.
- Data is summarized in map marker popups with summary of site data

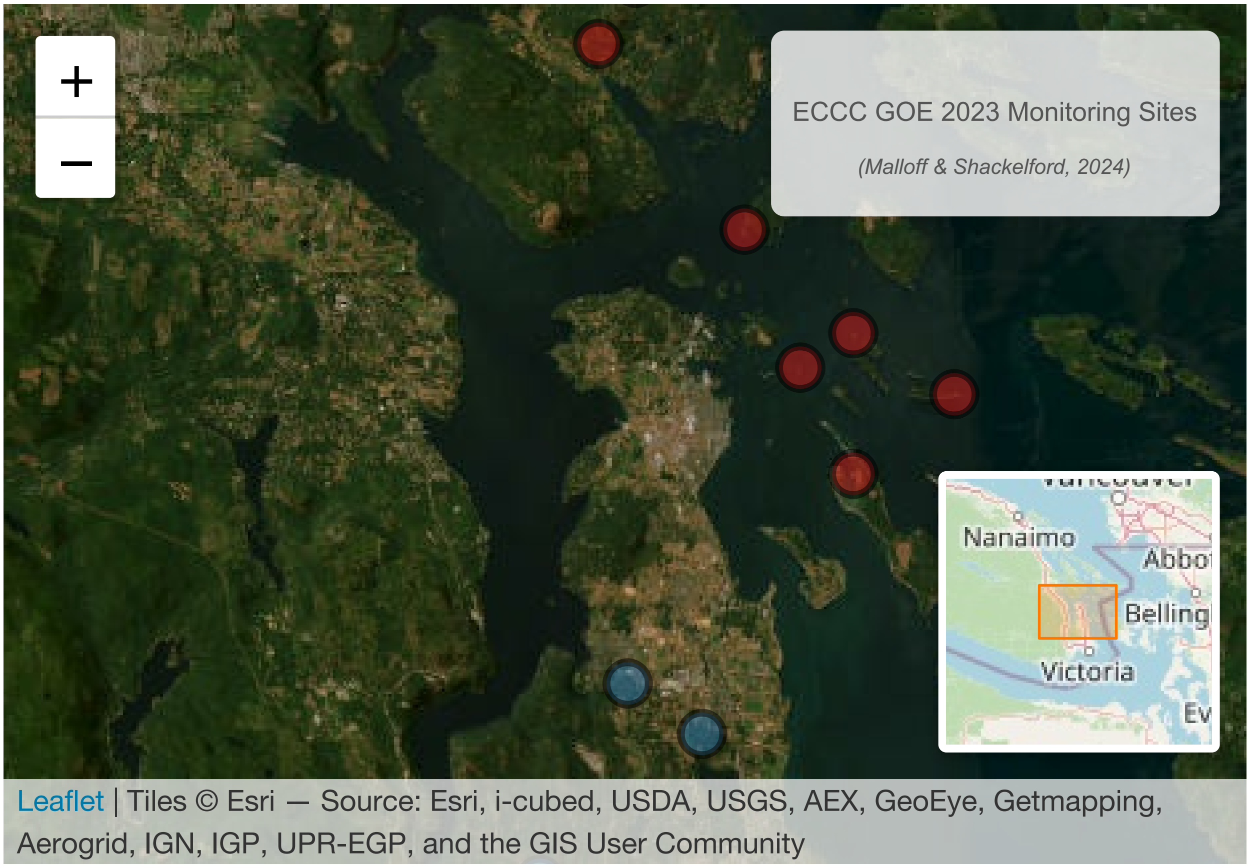

Figure 6: Interactive Leaflet Map with Example Sites Data

Subregion Plots

- These plot compare monitoring variable measurement results between two Subregions.

Violin/Bar Plot: Compare Subregions Exotic and Native Species

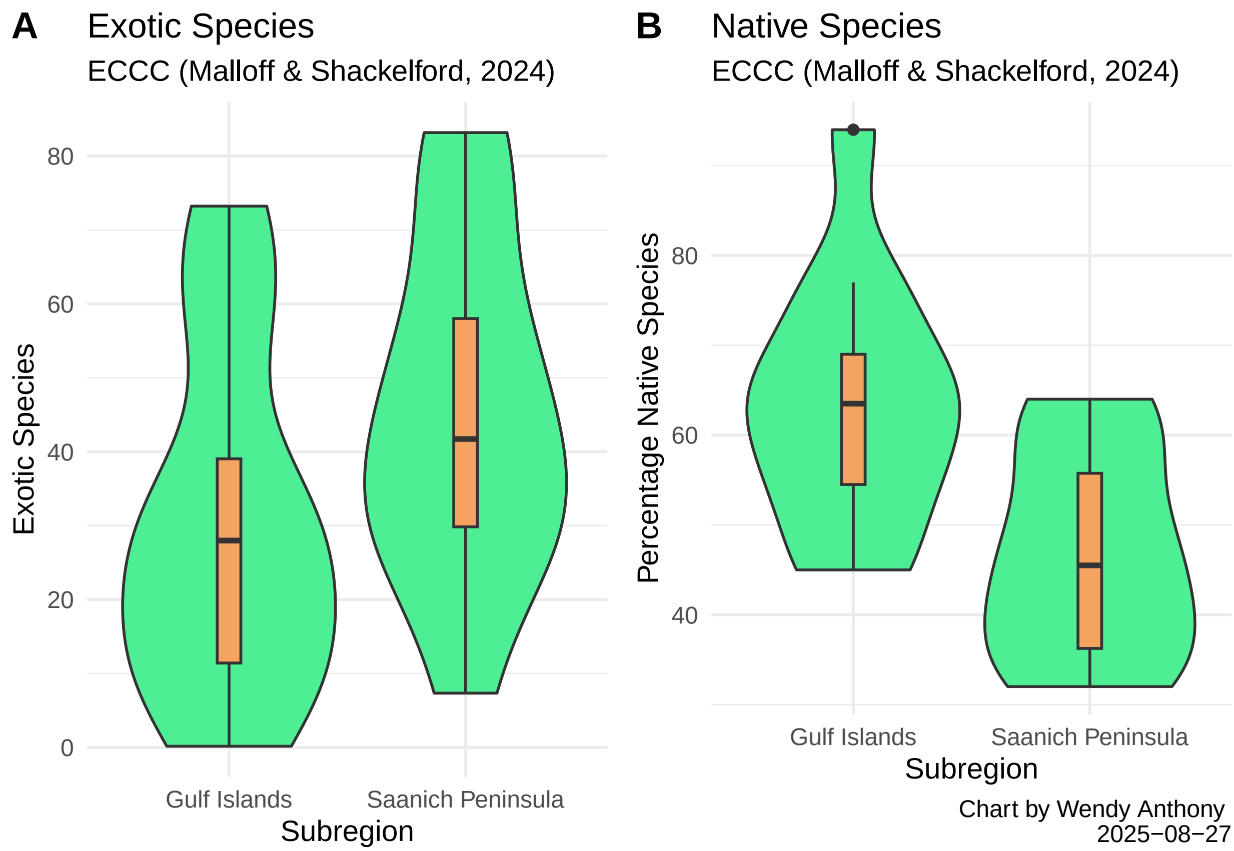

Figure 7: Violin / bar plot comparing variables of 2 Subregions

Site Plots

Point Plot: Compare Individual Sites by Restoration Intensity

- Sites are ordered by the level of restoration intensity, ranging from high to minimal
- This plotly-style chart becomes interactive by clicking data points for view a tool-tip of data results.

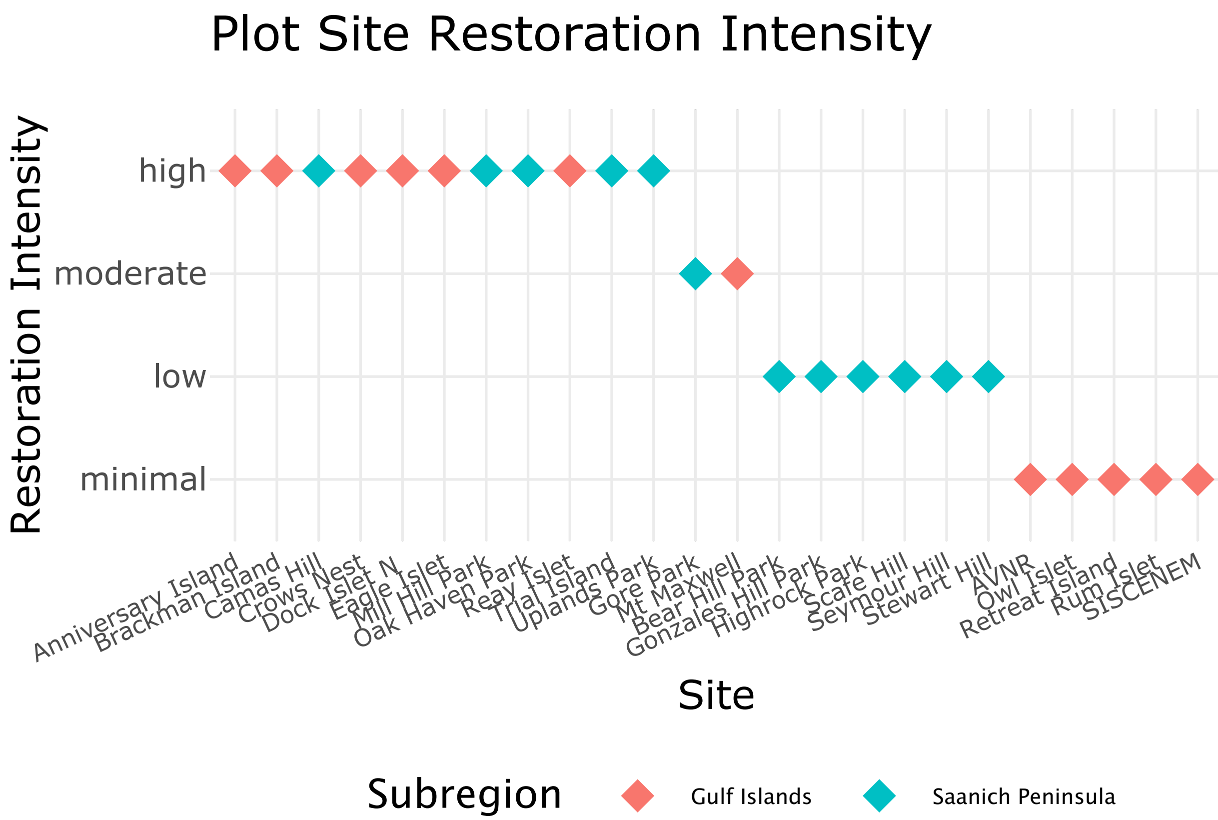

Figure 8: Point Plot of Restoration Intensity of Example Sites